

42115 Network Optimization

Network problems in liner shipping

Project Assignment, David Pisinger, DTU 2025

In this project assignment we will study how network optimization can be used to solve tactical and operational problems in liner shipping.

The assignment consists of five independent parts: In the first part we study how a irregular cargo can be packed on-board a vessel, and then we model a hub-and-spoke network. Next we identify bottlenecks in the network, and study the problem of repositioning empty containers. Finally, we study how a multi-commodity flow problem can be solved through column generation.

The project should be solved in **groups of 2-4 people**. The report may be written in Danish or English. The report should be uploaded as *one* pdf-file (no attachments) to DTU-inside. Questions 1–14 should be uploaded before **Monday 27 October at 8:00**. Questions 15–31 should be uploaded before **Sunday 7 December at 8:00**. You can use Julia instead of CPLEX, but in this case you should write the full LP-model in the report (not the Julia code).

It is only allowed to hand-in the project alone (group with 1 person), if you have strong arguments and get a written permission from the course responsible before 1st October.

If you are in doubt about what is expected as an answer to a question, ask at the exercises. **Questions will not be answered on e-mail** — instead come to the exercises.

Every year we see examples of plagiarism and other **exam cheating**. Such cases will be reported to the study administration who will take proper action. Please read the last page of this assignment for a clear overview of what is allowed.

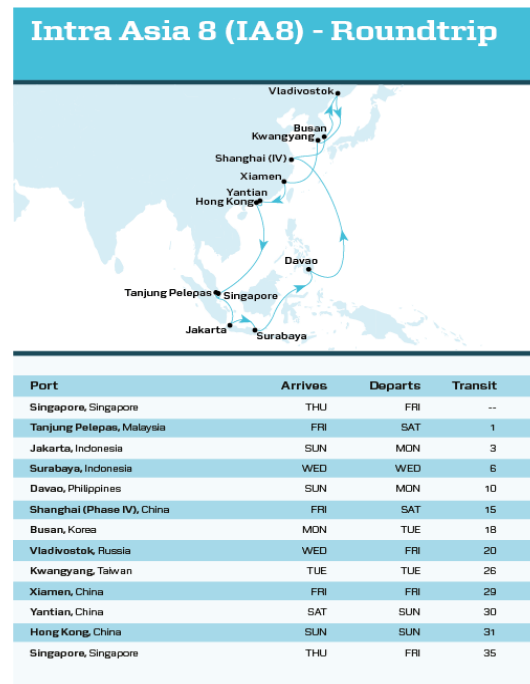
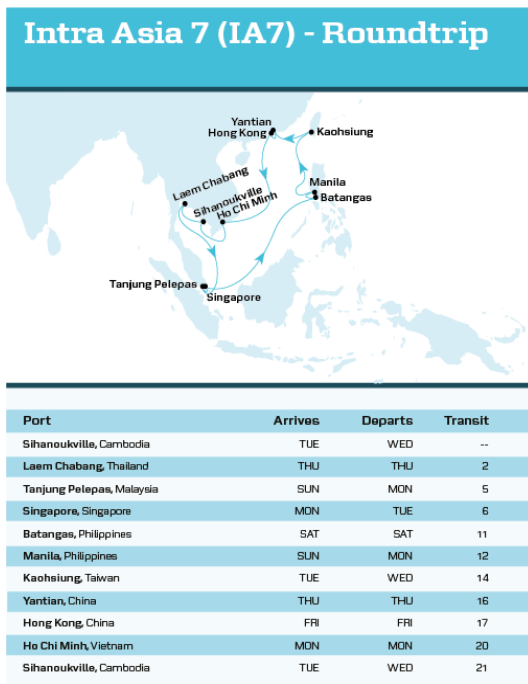


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Introduction

Containerized transportation is one of the most important inventions in logistics during the last century, giving us access to cheap and fast transportation of goods from all around the world. A modern vessel can carry more than 18.000 containers, resulting in low freight rates and a relatively small carbon footprint per container.

The vessels sail according to a route plan (like busses) as illustrated in the figures below. Weekly departure in each port is ensured by deploying additional vessels if the round trip takes more than 7 days. As an example, the IA7 round-trip takes 21 days, so three vessels are deployed to this route. In a similar way, five vessels are assigned to the IA8 round-trip as it takes 35 days.



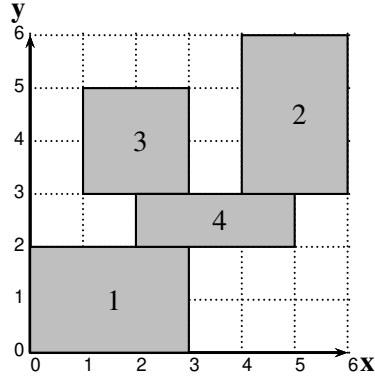
When goods are transported from one customer to another, it may be necessary to transship goods at a port. *Transshipment* means unloading the container from one vessel and storing it on the quay before it can be loaded to the next vessel. For instance, a container from Vladivostok to Ho-Chi-Minh will follow IA8 to Hong-Kong where it is transshipped to IA7 in order to be transported to the final destination.

Stowage of irregular cargo

Dangerous goods or irregular cargo is stowed separately on the deck as seen on the front page of this assignment. However, packing rectangles of different sizes can pose great challenges.

Mathematically, we can formulate the problem as a kind of two-dimensional bin packing problem. Given a set N of rectangles, each rectangle $i \in N$ has width w_i and height h_i . Rectangles may not be rotated. Furthermore, we are given a bin of size W times H . We wish to pack all rectangles within the bin, such that they do not overlap. Let (x_i, y_i) be the coordinates of the lower left corner of rectangle i .

The below example shows a set of rectangles, and a possible packing in the bin of size $(W, H) = (6, 6)$. In this packing we have $(x_1, y_1) = (0, 0)$, $(x_2, y_2) = (4, 3)$, $(x_3, y_3) = (1, 3)$, $(x_4, y_4) = (2, 2)$.



For every pair of rectangles i, j , where $i < j$, we introduce a relation $P_{ij} \in \{L, R, B, A\}$ defining their relative position. The meaning of the relations is:

$$\begin{aligned}
 P_{ij} = L &\implies x_i + w_i \leq x_j && (left) \\
 P_{ij} = R &\implies x_j + w_j \leq x_i && (right) \\
 P_{ij} = B &\implies y_i + h_i \leq y_j && (below) \\
 P_{ij} = A &\implies y_j + h_j \leq y_i && (above)
 \end{aligned}$$

If we make sure that at least one of $\{L, R, B, A\}$ holds between each pair of rectangles, then there is no overlap. The below table shows a possible choice of relations P_{ij} and the given dimensions (w_i, h_i) for the above problem

$i \backslash j$	1	2	3	4
1	-	L	B	B
2		-	R	A
3			-	A
4				-

i	1	2	3	4
w_i	3	2	2	3
h_i	2	3	2	1

Question 1 (3 point) Only considering the x -dimension, formulate the problem of finding the smallest legal x_i -coordinate for each $i \in N$ as a critical path problem. Describe the nodes, edges, and task durations. You can use the above instance as an example, but write the method in general terms. ■

Question 2 (1 point) Only considering the y -dimension, formulate the problem of finding the smallest legal y_i -coordinate for each $i \in N$ as a critical path problem. Describe the nodes, edges, and task durations. You can use the above instance as an example, but write the method in general terms. ■

Now, consider the following larger problem with $N = \{1, 2, 3, 4, 5\}$ rectangles. We have the following relations P_{ij} and dimensions (w_i, h_i) :

$i \backslash j$	1	2	3	4	5
1	-	L	B	A	R
2		-	B	R	R
3			-	A	R
4				-	B
5					-

i	1	2	3	4	5
w_i	2	1	4	4	1
h_i	3	4	1	2	4

Given a set N of rectangles, and the corresponding relations P_{ij} . Finding the smallest (x_i, y_i) legal coordinates for each rectangle i can be solved as a critical path problem.

Question 3 (3 point) For the above problem, find the smallest (x_i, y_i) coordinates for each rectangle $i \in N$ by solving two critical path problems (one for the x -dimension, and one for the y -dimension). Draw the resulting packing. Check whether all rectangles can be within $(W, H) = (5, 6)$. ■

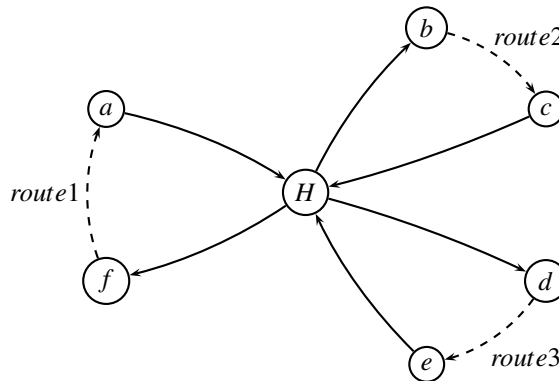
Question 4 (1 point) Not all relations P_{ij} lead to a logically meaningful packing. Show an example which does not represent a meaningful packing, and explain how it can be detected in Questions 1 and 2. ■

The above algorithm may be used to pack irregular items on-board a vessel. The algorithm will try all possible values of $P_{ij} \in \{L, R, B, A\}$ for all pairs of rectangles, each time finding the corresponding (x_i, y_i) coordinates of all the rectangles, and testing whether the packing is within the area $W \times H$. If a feasible solution is found, the algorithm terminates.

Modeling a hub port

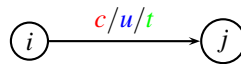
Many shipping companies operate with a *hub and spoke* model, where large ports act as a transfer point (hub). This is opposed to the *point to point* model, where direct connections are used to transfer containers between destinations. Hub and spoke networks make it possible to serve more destinations and decrease operational costs by using large vessels between the hubs. However, each hub impose some kind of delay in the transportation, and the operational costs of a hub may be quite large.

HUBLINE



Port a is the last port visited on route1 before visiting the hub H . Port f is the first port visited on route1 after visiting the hub H . Port c is the last port visited on route2 before visiting the hub H . Port b is the first port visited on route2 after visiting the hub H . Port e is the last port visited on route3 before visiting the hub H . Port d is the first port visited on route3 after visiting the hub H .

We will now look at how the hub port H in a liner shipping network, HUBLINE, can be modeled in graph terms as part of a larger network. Every edge can have an associated *cost* c , *capacity* u , and *time* t , as illustrated on the following edge (i, j)



The task is to model as many features as possible for the stated graphs.

We have three routes arriving to the hub as follows:

route	weekday
1	Saturday
2	Wednesday
3	Friday

Each vessel is only one day in the port (from 6:00 to 22:00), and it is assumed that all loading and unloading is done during this time. From the arrival weekdays, one can calculate how many days a transshipment will take.

The crane capacity u for loading/unloading is 10 containers per day for route1 and route2, while u is 7 containers per day for route3 since it is operated by a smaller crane. There are separate cranes for loading and unloading, having the same capacity of 10 (route1, route2) or 7 (route3).

The cost of a loading/unloading is 5€ for route 1 and 2, while it is 3€ for route 3. This means that if you transship from route 1 to route 3 you pay 5€ for unloading the container at H and 3€ for loading the container.

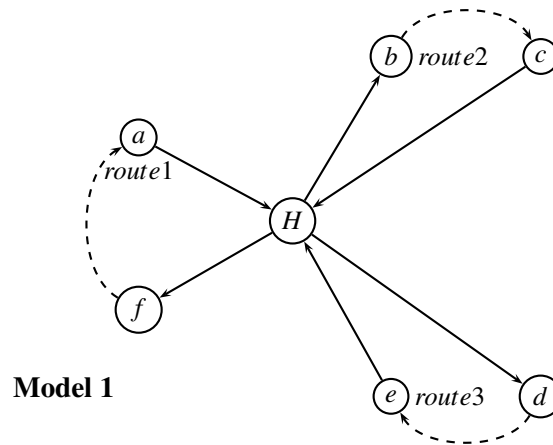
Finally, the port has a limited storage capacity of $\bar{u} = 20$ containers on shore. This means that each night the number of containers waiting for transshipment cannot exceed $\bar{u}$.

The shipping company must pay $\bar{c} = 2$ € for each container each night it is stored in the port. For instance, if a container arrives to the port Saturday and leaves Wednesday, then the storage cost is $4 \cdot 2 = 8$ €.

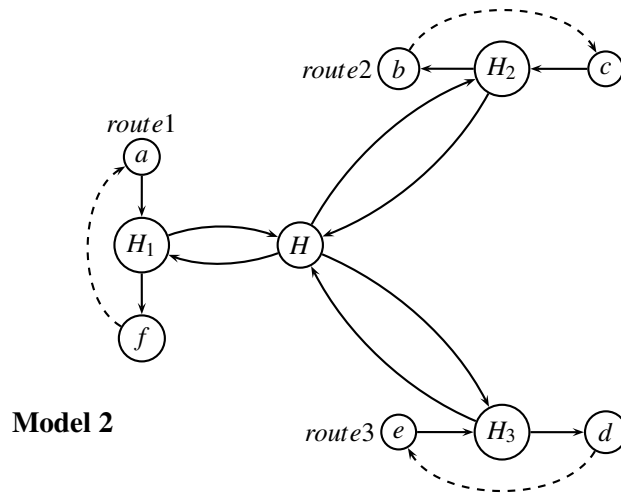
Example 1 A container from a to f will stay on board the vessel serving route1, and hence no costs will be paid, and no time spent in the port.

A container from a to b has to be unloaded from route1 (5€) and has to be loaded to route2 (5€), hence the crane costs are 10€. The transshipment takes 4 days, costing $4 \cdot 2 = 8$ € in storage. Hence, the total transshipment cost is 18€.

A container from a to d has to be unloaded from route1 (5€) and has to be loaded to route3 (3€), hence the crane costs are 8€. The transshipment takes 6 days costing $6 \cdot 2 = 12$ € in storage. Hence, the total transshipment cost is 20€. ■

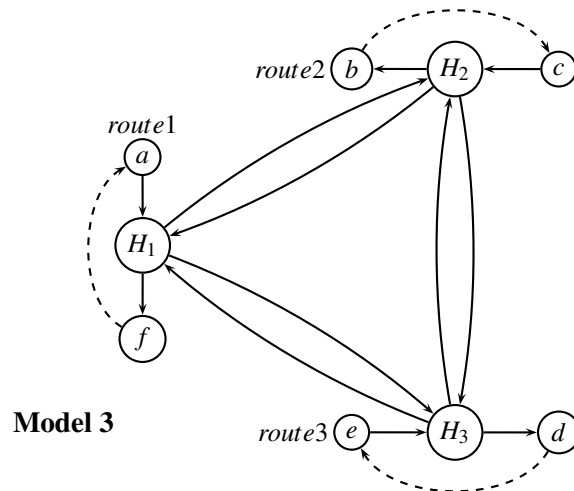


Model 1: The most intuitive way of modeling the hub port H is to use one node, that is visited by all three routes. However, it is not possible to assign transshipment costs or transshipment times to any of the edges. Neither is it possible to handle capacities. This is due to the fact that for each container in H we do not know whether the container stays on board the vessel, or whether it is transshipped to another route.



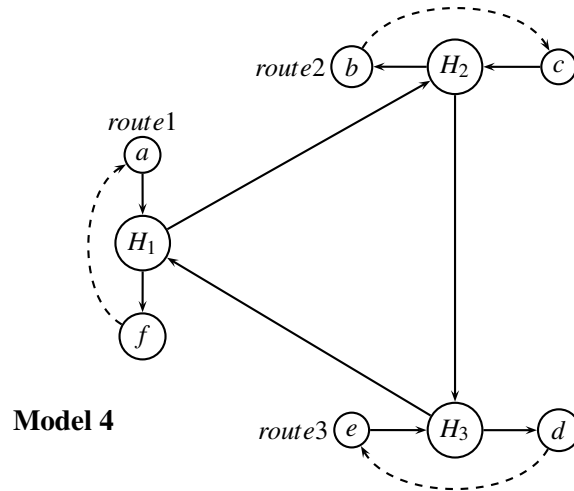
Question 5 (2 point) **Model 2:** Another possible way of modeling the hub port H is to use an extra node H_1 , H_2 and H_3 for each route. These are then connected to the node H .

Is it possible to model transshipment costs, transshipment time, crane capacities, storage capacities and storage costs? For those parameters where you answer yes, indicate the values on the edges. For those parameters where you answer no, argue why not. ■



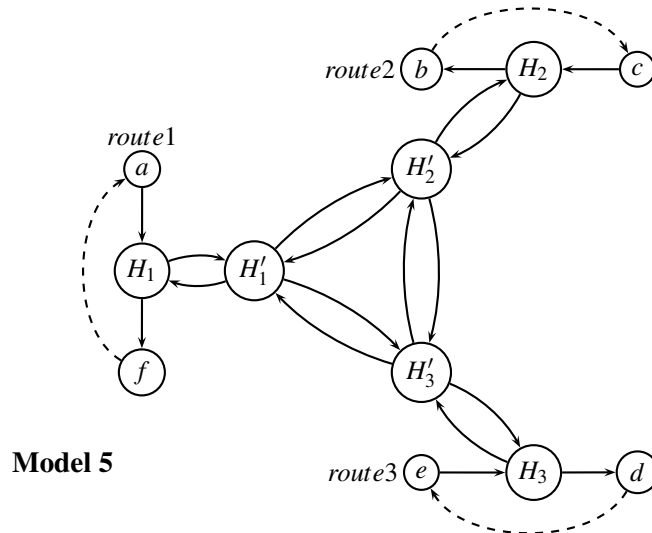
Question 6 (2 point) **Model 3:** Yet another possible way of modeling the hub port H is to use an extra node H_1 , H_2 and H_3 for each route. Edges are added *between all pairs of nodes* H_1 , H_2 , and H_3 .

Is it possible to model transshipment costs, transshipment time, crane capacities, storage capacities, and storage costs? For those parameters where you answer yes, indicate the values on the edges. For those parameters where you answer no, argue why not. ■



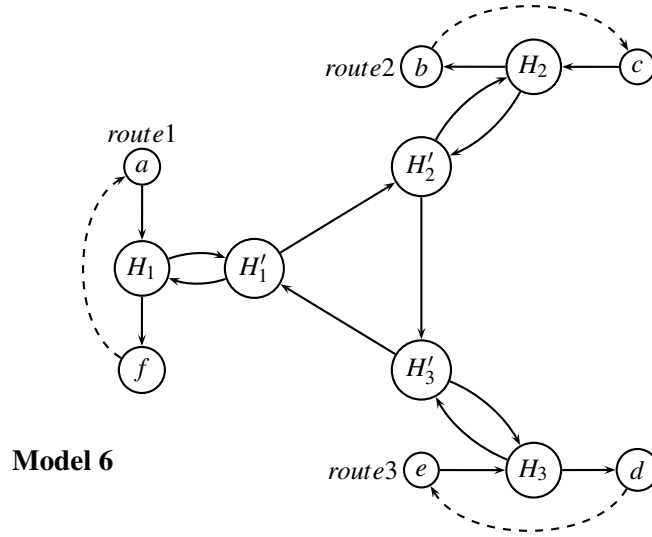
Question 7 (2 point) **Model 4:** An alternative to the above is a one-directional variant. To model hub port H we use an extra node H_1, H_2 and H_3 for each route. The nodes H_1, H_2 and H_3 are connected in a cycle ordered according to the time of arrival.

Is it possible to model transshipment costs, transshipment time, crane capacities, storage capacities, and storage costs? For those parameters where you answer yes, indicate the values on the edges. For those parameters where you answer no, argue why not. ■



Question 8 (2 point) **Model 5:** Next, we may model the hub port H by using an extra node H_1, H_2 and H_3 for each route, and additional nodes H'_1, H'_2 and H'_3 . Edges are added between all pairs of nodes H'_1, H'_2 , and H'_3 .

Is it possible to model transshipment costs, transshipment time, crane capacities, storage capacities, and storage costs? For those parameters where you answer yes, indicate the values on the edges. For those parameters where you answer no, argue why not. ■



Question 9 (2 point) **Model 6:** Finally, we may model the hub port H by using an extra node H_1 , H_2 and H_3 for each route, and additional nodes H'_1 , H'_2 and H'_3 . The latter nodes are connected in a cycle ordered according to the time of arrival.

Is it possible to model transshipment costs, transshipment time, crane capacities, storage capacities, and storage costs? For those parameters where you answer yes, indicate the values on the edges. For those parameters where you answer no, argue why not. ■

Question 10 (4 point) Assume that a hub port H has r routes arriving to the port, where r can be any positive number. For each of the six possible ways of modeling a hub, report in a table how many nodes and edges will be used to model the hub using big-O notation. You should not count the ports and edges on the r routes, since they need to be present in any case. We are only interested in studying the hub. Notice that in Model 3 all nodes H_i for $i = 1, \dots, r$ are mutually connected. Also in Model 5 all nodes H'_i for $i = 1, \dots, r$ are mutually connected.

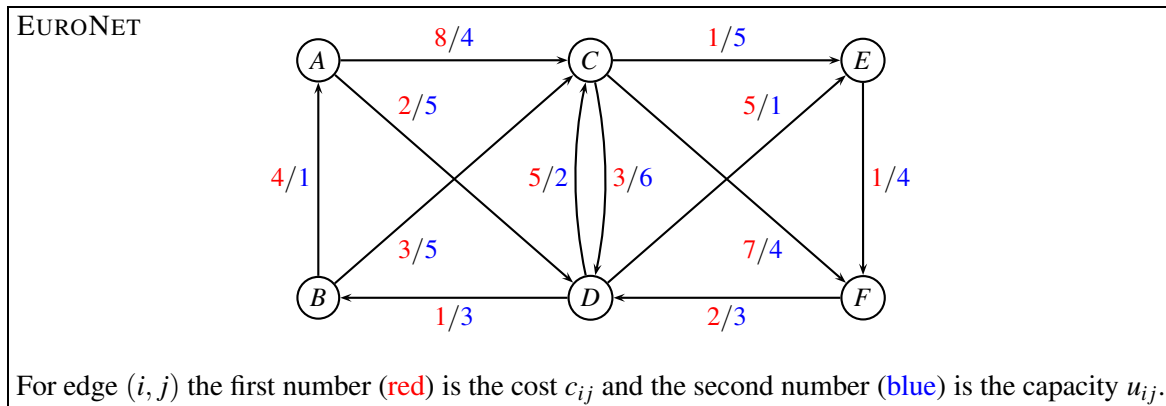
For each of the six models, extend the table to indicate which of the five criteria that can be handled.

Which model would you prefer to use if you need to model all five criteria and the number of routes r was large? ■

Finding bottlenecks in the network

The below instance, EURONET, shows a small liner shipping network. Nodes represent ports, and edges represent legs sailed by vessels. The capacities indicate the remaining capacity of the vessels after some mandatory orders have been fulfilled. Therefore, the capacity may vary for each leg.

In mathematical terms, we are given a weighted oriented graph $G = (V, E, c, u)$ where each edge $(i, j) \in E$ has an associated cost (pr. container) $c_{ij} \geq 0$, and a capacity $u_{ij} \geq 0$.



The shipping company is interested in finding an upper limit on the how much can be transported from a region S to another region T . (A *region* is a set of nodes).

Question 11 (2 point) Given a network $G = (V, E, c, u)$, with costs $c \geq 0$, and capacities $u \geq 0$. Argue which network algorithm that can be used to find the the largest amount of containers that can be sent from region S to T . ■

Question 12 (3 point) For the EURONET instance show how you find the capacity from region $S = \{A, B\}$ to region $T = \{E, F\}$. Give sufficient details to follow the calculations. You should use a network algorithm (not a MIP-solver). ■

The shipping company also wants to estimate the security of supply. Two paths between nodes a and b are *edge disjoint* if they do not share any edge. The more edge disjoint paths we have, the higher security of supply.

Question 13 (2 point) Given a network $G = (V, E, c, u)$, with costs $c \geq 0$, and capacities $u \geq 0$. Describe how you can find the maximum number of edge disjoint paths between region S and T as a network problem. ■

Question 14 (3 point) For the EURONET instance, calculate the maximum number of edge disjoint paths between region $S = \{A, D\}$ and region $T = \{C, F\}$. Give sufficient details to follow the calculations. You should use a network algorithm (not a MIP-solver). ■

_____ end of part 1 _____

Repositioning of empty containers

Due to the imbalance in trade, empty containers tend to accumulate at various ports. For instance China is shipping much more goods than Europe, resulting in a surplus of empty containers in Europe and a demand of empty containers in China.

For EURONET, one could have the following inventory of empty containers:

node	A	B	C	D	E	F
inventory	7	9	5	4	2	3

It is assumed that all empty containers are equal. We want to reposition empty containers such that we have the same number of empty containers at each node.

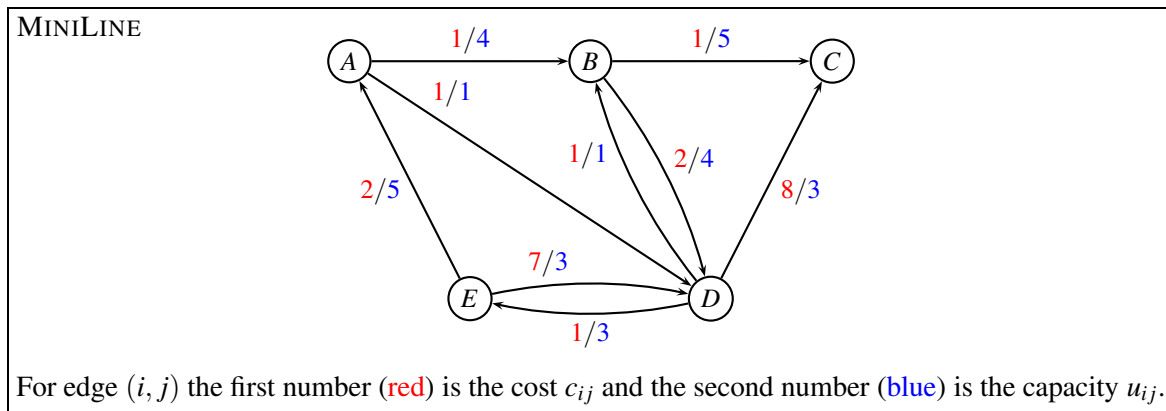
Question 15 (3 point) Formulate the repositioning of empty containers as a *minimum cost flow problem*. Define the graph and the demands in each node. ■

Question 16 (4 point) Solve (by hand) the EURONET repositioning problem, using a network algorithm (not a MIP-solver). State which algorithm you use, and give sufficient details to make it possible to follow your calculations. Report the optimal solution. ■

Flowing containers

The below instance, MINILINE, shows a small liner shipping network. Nodes represent ports, and edges represent legs sailed by vessels. The capacities indicate the remaining capacity of the vessels after some mandatory orders have been fulfilled. Therefore, the capacity may vary for each leg.

In mathematical terms, we are given a weighted oriented graph $G = (V, E, c, u)$ where each edge $(i, j) \in E$ has an associated cost (pr. container) $c_{ij} \geq 0$, and a capacity $u_{ij} \geq 0$.



Frequently there are several different ways of transporting containers, hence the task of the shipping company is to utilize the network in the best possible way, respecting the capacity of the container vessels.

Customers prefer to have all their containers sent along the same route, as this makes the logistics easier. Hence, we assume that all flows must be *unsplittable*, i.e. for each commodity $k \in K$ the d_k units must follow the same route.

In mathematical terms, we are given a weighted oriented graph $G = (V, E, c, u)$ where each edge $(i, j) \in E$ has an associated cost $c_{ij} \geq 0$ and a capacity $u_{ij} \geq 0$. Moreover we are given a number

of commodities $K = \{1, \dots, m\}$ each given by a triple (s_k, t_k, d_k) , where s_k denotes the source of commodity k , t_k denotes the target of commodity k , and d_k denotes how many units to be sent. Using decision variables x_{ij}^k to indicate whether or not commodity $k \in K$ is using edge $(i, j) \in E$, we get the following formulation of the *unsplittable multicommodity flow problem*

$$\min \sum_{k \in K} \sum_{(i,j) \in E} c_{ij} d_k x_{ij}^k \quad (1)$$

$$\text{s.t.} \quad \sum_{i \in V} x_{ij}^k - \sum_{i \in V} x_{ji}^k = 0 \quad k \in K, j \in V \setminus \{s_k, t_k\} \quad (2)$$

$$\sum_{i \in V} x_{ij}^k - \sum_{i \in V} x_{ji}^k = -1 \quad k \in K, j = s_k \quad (3)$$

$$\sum_{i \in V} x_{ij}^k - \sum_{i \in V} x_{ji}^k = 1 \quad k \in K, j = t_k \quad (4)$$

$$\sum_{k \in K} d_k x_{ij}^k \leq u_{ij} \quad (i, j) \in E \quad (5)$$

$$x_{ij}^k \in \{0, 1\} \quad k \in K, (i, j) \in E \quad (6)$$

The first constraint (2) ensures flow conservation. The next constraint (3) ensures for each k that the commodity flows out from s_k . The following constraint (4) ensures for each k that the commodity flows into t_k . Finally, the capacity of each edge needs to be respected by (5). Constraint (6) ensures that all d_k containers follow the same route. We will call this formulation the *edge formulation* of the *unsplittable multicommodity flow problem*. The unsplittable multicommodity flow problem is $\mathcal{NP}$ -hard, as shown by Khan (2000), meaning that the problem is very difficult to solve for large-sized instances.

We consider again the route net MINILINE where we now have the demands

k	s_k	t_k	d_k
1	E	D	2
2	D	C	2
3	E	C	1

Commodity $k = 1$ is shipping two containers from E to D , commodity $k = 2$ is shipping two containers from D to C , and commodity $k = 3$ is shipping one container from E to C .

Question 17 (3 point) Write up the edge formulation for MINILINE and solve it as an LP-problem by use of CPLEX. Enclose the CPLEX model and CPLEX solution. Report the selected route of each container in a table. ■

Question 18 (1 point) Solve the edge formulation for MINILINE as an IP-problem. Enclose the CPLEX model and CPLEX solution. Report the selected route of each container in a table. ■

Question 19 (1 point) Compare the IP and LP solution, and discuss whether the LP-relaxation provides a good lower bound. (As a rule of thumb, less than 1% is an excellent bound, less than 5% is a good bound, between 5 and 20% is a usable bound.) ■

Question 20 (3 point) For each commodity k in MINILINE, write up a table with all possible simple routes from s_k to t_k through the graph $G = (V, E, c, u)$. A route may only use edges (i, j) where the capacity u_{ij} is at least d_k units. For each commodity k and each feasible route, state the corresponding cost of sending the d_k units. (**hint:** There should be 10 distinct routes, since no cycles are allowed.) ■

Let R_k denote the set of possible routes from s_k to t_k allowing d_k units of capacity. Let moreover $a_{r,ij}^k \in \{0,1\}$ denote if route r for commodity k is using edge (i,j) , and let $\hat{c}_r^k$ denote the cost for sending d_k containers along route $r \in R_k$. Let x_r^k for each route $r \in R_k$ denote whether or not this route is used for commodity k . We can now write up an alternative formulation of the problem, which we will call the *path formulation* of the *unsplittable multicommodity flow problem*.

$$\min \sum_{k \in K} \sum_{r \in R_k} \hat{c}_r^k x_r^k \quad (7)$$

$$\text{s.t.} \quad \sum_{k \in K} \sum_{r \in R_k} d_k a_{r,ij}^k x_r^k \leq u_{ij} \quad (i,j) \in E \quad (8)$$

$$\sum_{r \in R_k} x_r^k = 1 \quad k \in K \quad (9)$$

$$x_r^k \in \{0,1\} \quad k \in K, r \in R_k \quad (10)$$

Constraint (8) ensures that the capacity of each edge is respected, while (9) ensures that at least one route from s_k to t_k is selected. Finally, (10) ensures that the demand d_k is not split. We use greater-equal signs in (9), as they are easier to handle for the LP-solver. Since $\hat{c}_r^k > 0$, an optimal solution will never use more than one route for each commodity.

Question 21 (3 point) Write up the path formulation of MINILINE. You can use CPLEX format. ■

Question 22 (1 point) Solve the path formulation of MINILINE to LP-optimality with CPLEX and report the lower bound. Enclose the model. ■

Question 23 (1 point) Solve the path formulation of MINILINE to integer optimality with CPLEX and report the solution value. Enclose the model. ■

Question 24 (1 point) Compare the IP and LP solution, and discuss whether the LP-relaxation provides a good lower bound. ■

A disadvantage of the path formulation is that many routes from s_k to t_k may exist for each commodity, such that the path formulation in practice becomes unsolvable due to the large number of decision variables.

If we only are interested in solving the LP-relaxed problem, this problem can be avoided by writing up the routes only when they are needed in the simplex algorithm, hoping that just a small subset of the routes will be generated. This approach is known as *column generation*.

To illustrate the principle, consider again the LP-relaxed path formulation of MINILINE, in which we only consider the following routes:

commodity k	route	cost
1	ED	14
2	DC	16
3	EDC	15

This leads to the following LP-model corresponding to (7) - (10):

$$\begin{array}{llllll}
\min & 14x_1^1 & + & 16x_1^2 & + & 15x_1^3 \\
\text{s.t.} & & & & & \\
& & & & \leq & 4 & (y_{AB}) \\
& & & & \leq & 1 & (y_{AD}) \\
& & & & \leq & 5 & (y_{BC}) \\
& & & & \leq & 4 & (y_{BD}) \\
& & & & \leq & 1 & (y_{DB}) \\
& & 2x_1^2 & + & x_1^3 & \leq & 3 & (y_{DC}) \\
& & & & \leq & 3 & (y_{DE}) \\
& 2x_1^1 & & + & x_1^3 & \leq & 5 & (y_{EA}) \\
& & & & \leq & 3 & (y_{ED}) \\
& x_1^1 & & & & \geq & 1 & (\bar{y}_1) \\
& & x_1^2 & & & \geq & 1 & (\bar{y}_2) \\
& & & x_1^3 & & \geq & 1 & (\bar{y}_3) \\
& 0 \leq x_1^1, x_1^2, x_1^3 \leq 1 & & & & & &
\end{array} \tag{11}$$

We call the above model, the *restricted master problem*.

Let y_{ij} denote the dual variable corresponding to constraint (8) for edge (i, j) , and let $\bar{y}_k$ denote the dual variable corresponding to constraint (9) for commodity k . The dual variables can be interpreted as follows: Variable $-y_{ij}$ says how much it costs to use edge (i, j) . Variable $\bar{y}_k$ says how much is saved by finding an alternative route from s_k to t_k . The dual variables for the above model (11) are $y_{ij} = 0$ for all $(i, j) \in E$, while $\bar{y}_1 = 14$, $\bar{y}_2 = 16$, $\bar{y}_3 = 15$.

Question 25 (3 point) Write up an equation stating how the reduced cost of a route r can be calculated. Next, calculate the reduced cost of all routes from Question 20 given the above dual variables. Indicate the route $r \in R_1 \cup \dots \cup R_m$, having the most negative reduced cost. ■

Since there can be an exponential number of routes for a given problem, calculating the reduced cost of all routes may be very time consuming, so we will use a more efficient approach to find the route with most negative reduced cost.

Question 26 (2 point) For a given commodity k describe how a route with most negative reduced cost can be found using a shortest path algorithm. Describe how the edge costs can be calculated, and discuss whether they can become negative. Which algorithm can you use to solve the shortest path problem? ■

Question 27 (2 point) Draw the network for each commodity k (nodes, edges, costs, start-node, end-node), and report the optimal solution to the shortest-path problem as well as the reduced cost. (you do not need to provide details about how the shortest-path problem was solved). ■

Question 28 (1 point) Extend the restricted master problem (11) with the new route. Solve the new problem to LP-optimality with CPLEX and report the dual variables. ■

Question 29 (5 point) Repeat the process of adding new promising routes to the restricted master model until no more routes with negative reduced cost can be added to the model. For each iteration, find the route with most negative reduced cost, using a shortest path algorithm. In each iteration write

up the restricted master problem, and report the primal and dual variables as well as the objective. (**Hint:** You do not need to draw the graphs, just describe in words which modifications were made to the graph. Also, you do not need to describe how the shortest path problem was solved. Just report the shortest path and its cost.) ■

Question 30 (1 point) Discuss how many routes were added in the column generation approach and compare it to the total number of routes. ■

Question 31 (5 point) **Theoretical question** Prove that (in general) you will find tighter bounds by solving the LP-relaxation of the path formulation than solving the LP-relaxation of the edge formulation. (**Hint:** you need to show that any feasible solution to the LP-relaxed path formulation also is a feasible solution to the LP-relaxed edge formulation. Moreover you should show that at least one instance exists for which the path formulation gives tighter bounds than the edge formulation.) ■

_____ end of part 2 _____

Tips and tricks

Binary variables

In CPLEX you declare binary variables with the command `integer`, while you declare integer variables with the command `general`.

Bounds

Restricted master problem: Avoid using upper bounds on variables, since they are handled implicitly in Simplex, and hence the dual variables will *not* be reported. You can formulate upper bounds as constraints, or you can drop the bound if you know the variable will never exceed the bound.

Checking your calculations in column generation

Assume that all columns are known in advance, but only a subset of the columns are used in the restricted master problem. Add a constraint

$$x_r^k \leq 0 \quad (12)$$

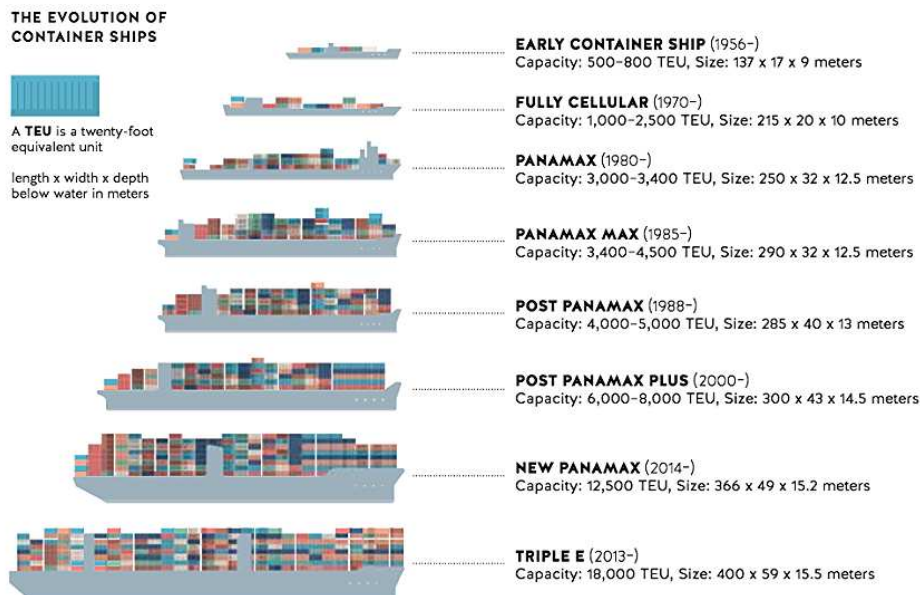
to all columns which are not used. This will “inactivate” the columns, and the same input file can be used for several iterations with only small modifications. The dual variable corresponding to (12) is actually the reduced cost of the column and can be used to check your calculations.

Output from CPLEX

If you want to use the output from CPLEX in your report, you can mark the text on the screen, and copy it into your document. Alternatively, you can use the log file from CPLEX. It can be found in the same directory as your input file. CPLEX writes all output to the log file.

Print model in Julia

There are commands `show(model)` and `latex_formulation(model)` to print the full model, so you can include it in your report.



DTU's code of honour

DTU's code of honour says:

As a student at DTU I acknowledge DTU's code of honour for examinations and other academic activity. I accept and respect that the high standards for accountability and academic and scientific integrity apply to me as a student just as they apply to lecturers, researchers, and students at the university.

I therefore show responsibility and integrity in my work, and my exam submissions always reflect my own work, without having received inappropriate oral or written assistance, including digital sources. I know that I must never copy (plagiarize) other people's ideas, thoughts, reports or articles, but I am permitted to quote and refer to them using quotation marks and source references. I also know that I am not allowed to communicate with others during a written exam.

For this project assignment (which is part of the exam) it means:

- You may discuss solution methods with other groups
- You may get hints from students who followed the course previous years
- You can ask TA's to clarify the questions, and possibly give you a hint if you are stuck
- You may use figures from the project assignment handed out

However,

- Every group should write their own report and develop their own models/code and run their own experiments
- You may not exchange copies of your report or models between the groups
- You may not use code or models from the internet or from other students (even if you change variable names, or reformat the code)
- You may not read project assignments or solutions from previous years of Network Optimization (even if you only use them for inspiration)